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Section 08

Project:

**SpaceSync: Real-Time Campus Study Spot &
Facility Manager**

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1.Introduction:

The rapid increase in digital technologies has greatly changed university operations and ways in which services are provided. Campuses continue to expand and enroll more students, making the effective management of physical learning spaces a complex task. Too often, students waste much time searching for available study rooms, seats in the library, or discussion spaces, which causes overcrowding and raises their stress levels. Such challenges make it necessarily important to give a wake-up call toward smarter, integrated campus facility management.

The goal of this Design Thinking project is to propose SpaceSync: Real-Time Campus Study Spot & Facility Manager, a future digital campus system that is aimed at enhancing study space and facility utilization and accessibility. It focuses on user-centered design through understanding the real needs of students and staff by offering an intuitive interface that enables real-time monitoring, booking, and management of campus study spots. This project applies the process of Design Thinking-empathy, define, ideate, prototype, and test-in enhancing learning efficiency, reducing stress, and supporting a more organized campus environment.

2. Design Thinking Process:

2.1 Empathy (*Mashrur*)

- Interview Transcript (Simulated):

Interviewer (Mashrur): Could you explain how you usually go about finding a place to study on campus?

Participant (Sarah): To be honest, it's annoying. After my morning class, I usually go straight to the library (PSZ). Level 3 is usually crowded when I walk up to it. I have to search every aisle for a vacant seat close to a power outlet. After ten minutes, if I still can't find one, I give up and walk all the way to the Student Union (SUB) in the hopes that there will be room there.

Interviewer (Mashrur): How do you feel about that process?

Participant (Sarah): It wears me out and makes me nervous. I feel like I'm squandering important study time by being a "nomad" who wanders around. I'm sweaty and irritated by the time I find a seat, and I'm no longer motivated to study. When everyone is vying for space during exam weeks, it gets even worse.

Interviewer (Mashrur): Have you ever attempted to reserve a lab or discussion room in advance?

Participant (Sarah): "I tried once, but the system was confusing." I was unsure whether the room was genuinely available or if the system had simply not been updated. Additionally, I occasionally walk all the way to a "booked" room only to discover that it is empty because the person did not show up, but I am unable to use it because I did not reserve it. It seems like a waste of money.

Interviewer (Mashrur): How would you solve this issue if you had a magic wand?

Participant (Sarah): I simply want to launch an app on my phone that displays a map, similar to a movie theater seating chart, that pinpoints the precise tables that are currently green (empty). In order to be certain that I have a seat waiting for me, I would like to be able to "chope" (reserve) a particular seat for fifteen minutes

while I'm walking there.

- User persona (age, role, background, pain points):
 - **Name:** Sarah
 - **Role:** 3rd Year Software Engineering Student
 - **Background:** Lives off-campus (Kolej Perdana), relies on campus facilities during 3-hour gaps between classes.
 - **Current Behavior:** physically walks from building to building (library to student union) to find study spots.
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- Empathy insights

Uncertainty: The user suffers from anxiety because they never know if a facility is available until they physically arrive.

Physical Fatigue: The current method of finding space requires excessive walking, which lowers energy for actual studying.

Inefficiency: The user perceives the "manual search" for seats as a direct loss of academic productivity.

2.2 Define (*Youssef*)

Problem Statement (Point of View - POV):

Sarah, a 3rd-year Software Engineering student living off-campus ("The Nomad"), **needs to guarantee** a quiet study spot with a power outlet immediately after her class ends, **because** the uncertainty of finding a seat causes her physical anxiety and wasting 20 minutes "hunting" for a table makes her feel like she is losing critical productivity time.

Needs as Verbs:

- To **visualize** seat availability before walking to the building.
- To **reserve** ("chope") a specific spot to eliminate the race against other students.
- To **optimize** her schedule by avoiding overcrowded areas during peak hours.

Reframing the Problem (How Might We):

Instead of asking "How can we build more study rooms?", we reframed the challenge to:

"How might we make the existing invisible vacancy of campus facilities visible and actionable for students like Sarah in real-time?"

2.3 Ideate (*Youssef*)

Brainstorming Process:

We utilized the "Divergent Thinking" method, focusing on quantity over quality initially to generate a wide range of solutions for Sarah's anxiety and time-wasting issues. We looked at the problem from three angles: Physical changes, Policy changes, and Technological solutions.

Idea Selection Explanation (Convergent Thinking):

After generating multiple concepts (detailed in Section 5.3), we selected Idea #1: The IoT Sensor & Booking App (SpaceSync).

We chose this solution because:

1. Feasibility: It uses existing infrastructure (Wi-Fi/Sensors) rather than building new rooms.
2. User-Centric: It directly solves Sarah's "Uncertainty" pain point by offering a "Movie Theater" style map view.
3. Scalability: It solves the staff's management issue simultaneously by generating usage data.

2.4 Prototype (*Fares*)

Interface-only prototype description:

SpaceSync was designed and developed as a non-functional, high-fidelity visualization of the proposed system. No coding was required; the focus remained entirely on User Interface (UI) and User Experience (UX) design. The primary objective was to create a clean, intuitive interface that allows students and staff to access sensor data easily. This visualization ensures users can identify available spaces in real-time for booking and data analysis. The design utilizes visual cues, such as color-coding (green for available, red for occupied), to facilitate efficient decision-making for students.

Screens Flow explanations:

1- Login screen:

Users enter their unique UTMID to securely authenticate, gain authorized access, and verify student identity.

2- Home/Dashboard Screen:

It provides a graphical overview of the usage of facilities on charts, with quick-action icons for Booking, Finding Spots, Reporting, and Profile. It also provides an embedded map for locating the nearest quiet places to study or particular facilities based on real-time locations.

3- Real-time Tables view or Map:

It also gives an integrated view of the sensor readings to form the live map of the library or 24-hour study room. The occupancy level is indicated with a high level of accuracy ranging from 86% to 97% depending on the efficiency of the algorithm used.

4- Booking confirmation:

The confirmation page verifies the reservation. Although it recognizes that there could be a slight delay with the sensors, this page can be considered digital proof of the reservation history for the user.

Tools Used in Building this prototype:

The use of Canva for prototyping was based on its collaboration capabilities through cloud storage, which ensures data security and easy collaboration without limitations due to hardware factors on desktop computers.

2.5 Test (*Fares*)

Testing Process:

We conducted a Cognitive Walkthrough with the prototype, asking users to perform tasks such as finding a vacant study space and determining how full the library seemed to be. The group observed users' interaction with the layout of the interface while recording their verbal feedback to determine where users were having trouble with navigation or where functionality was lacking.

Who Tested it:

- User A (Student, FC, Year 3): Frequently studies on campus but struggles to find available seats.
- User B (Staff, FC): Unfamiliar with IoT sensor systems.
- User C (Library Staff): Expert on library facility usage.

Feedback Received:

On the whole, participants felt that the system was innovative and useful. They did find that there was a little too much empty space at certain points upon opening the interface. They suggested adding functional information such as opening and closing times and implementing a recommendation system powered by AI to optimize search results.

Improvement Suggestions:

1. **Operational Details:** Display opening and closing hours for selected facilities on the map.
2. **Resource Integration:** Add links to other UTM resources to utilize the empty screen space.
3. **AI Recommendation:** Implement an AI chatbot (via Google AI Studio API) to recommend study spots based on user queries.
4. **Admin Analytics:** Develop a "Staff View" dashboard using Machine Learning (ML) to predict peak times, aiding in resource management decisions.
5. **Ecosystem Integration:** Include quick links to other UTM applications for a unified student experience

3. Problem, Solution & Team Working:

3.1 Problem Description:

As it stands on campus today, searching for a good study spot has become a problem for students and university employees. Study rooms are filled quickly, library seating is nowhere to be found, computer labs are crowded out, group areas are taken while other spots sit vacant. Without a system showing available study spots in real time, people have to search around campus.

On the other hand, the campus facilities team finds it hard to track usage patterns, identify rush hours, and allocate upkeep schedules accordingly. Such unclear data leads to clouded decision-making, with resources being inefficiently utilized. This consequently affects students' focus in class, decreases staff efficiency, and further paints the whole campus life in a poor light.

3.2 Proposed Solution: SpaceSync System

To overcome these challenges, we offer a solution called SpaceSync, a real-time study and facility management system for campuses. The system is a one-stop digital platform designed to enable users to view instantly available study space, room booking, and notifications in a very simple interface.

Main features of SpaceSync:

A live display of where library study spaces are available or being used

- Internet booking and reservation system for study rooms and amenities

- Notifications and reminders related to booking

- Employee management dashboards for the purpose of smarter decision-making Combined, these elements are meant to reduce overcrowding as well as promote a well-efficient space organization system for a more productive and stress-free learning environment. SpaceSync's integration with the Future Digital Campus vision incorporates technology with ease of use and creative problem-solving.

3.3 Team Working and Collaboration

The success of this project is based on effective teamwork and collaboration among all group members. Each member was assigned specific responsibilities aligned with their strengths and the Design Thinking phases. Regular discussions and idea-sharing sessions were conducted to ensure consistency and alignment throughout the project.

As the team leader, I coordinated the project workflow, ensured that deadlines were met, and integrated contributions from all members into a coherent final output. Collaborative decision-making was practiced during the ideation and prototype phases to ensure that the proposed solution addressed real user needs. This teamwork approach allowed the group to produce a comprehensive and well-structured design thinking project.

4. Design Thinking Assessment Points:

Design thinking assessment is what tethers the solution to what a user actually needs and to the goals of the project. In this project, we carried out assessments at two key moments to check progress and raise the quality of what we're building.

First, we stopped to reflect at the transitions between phases of design thinking. At each transition—after Empathy, after Define, after Ideate, after Prototype—the team looked back and reflected on what was learned and created before advancing to the next step. For example, we questioned whether the insights coming out of the empathy work really represented the user problems before we defined the core challenges. This helped us identify and fill gaps, tighten ideas, and make improvements early.

Secondly, the assessment happened during the final project demo. Here we saw altogether the solution, the flow of the prototype, and user experience in order to judge whether SpaceSync truly deals with the identified problems. We considered feedback from teammates and simulated users for usability, clarity, and practicality. These checkpoints ensured the final proposal was user-centered, feasible, and aligns within the Future Digital Campus theme.

5. Design Thinking Evidence:

5.1 Empathy Evidence (*Mashrur*)

- Interview table

Question	User Response (Simulated)	Key Insight / Pain Point
Could you explain how you usually go about finding a place to study on campus?	To be honest, it's annoying. After my morning class, I usually go straight to the library (PSZ). Level 3 is usually crowded when I walk up to it. I have to search every aisle for a vacant seat close to a power outlet. After ten minutes, if I still can't find one, I give up and walk all the way to the Student Union (SUB) in the hopes that there will be room there.	Physical Fatigue: The current process requires excessive physical movement and wasted time just to find a simple amenity.
How do you feel about that process?	It wears me out and makes me nervous. I feel like I'm squandering important study time by being a "nomad" who wanders around. I'm sweaty and irritated by the time I find a seat, and I'm no longer motivated to study. When everyone is vying for space during exam weeks, it gets even worse.	Productivity Loss: The search for space negatively impacts mental focus and academic performance.

Have you ever attempted to reserve a lab or discussion room in advance?	"I tried once, but the system was confusing." I was unsure whether the room was genuinely available or if the system had simply not been updated. Additionally, I occasionally walk all the way to a "booked" room only to discover that it is empty because the person did not show up, but I am unable to use it because I did not reserve it. It seems like a waste of money.	Uncertainty: Users do not trust the current manual/legacy systems due to lack of real-time data verification.
How would you solve this issue if you had a magic wand?	I simply want to launch an app on my phone that displays a map, similar to a movie theater seating chart, that pinpoints the precise tables that are currently green (empty). In order to be certain that I have a seat waiting for me, I would like to be able to "chope" (reserve) a particular seat for fifteen minutes while I'm walking there.	Need for Real-Time Data: Users prioritize immediate visual confirmation of availability over general schedules.

- Persona summary

Profile Attribute	Description
Name	Sarah
Role	Year 3 Software Engineering Student
Demographics	21 years old, lives off-campus (Kolej Perdana).
Goals	To maximize her 3-hour gap between classes for coding assignments; requires a quiet space with reliable power outlets and Wi-Fi.
Behaviors	Carries a heavy laptop bag; visits multiple locations (Library, Labs, SUB) in a single day searching for space; hesitates to ask strangers if a seat is taken.
Frustrations	• Wasted Time: Spending 20+ minutes walking between buildings to find a spot. • Anxiety: Fear of running out of laptop battery because she cannot find a seat near a plug. • Uncertainty: Not knowing if a "booked" room is actually occupied or abandoned.
Quote	"I feel like a nomad wandering around campus. I just want to know where an empty seat is before I start walking."

5.2 Define Evidence (Youssef)

- Problem statements

The 5-Whys Analysis:

- *What makes Sarah anxious? Because she is worried about finding a place where she can study.*
- *- Why can't she determine whether she is to find one? "Because the occupancy information is not on the web; it is only accessible when you are there."*
- *- Why is it only visible physically? It is because there are no sensors on campus that provide real-time data.*
- *- Why is that the problem? It is because she ends up wasting energy by walking from building to building.*
- *Root Cause: The lack of digital transparency for physical assets.*

HMW Questions Generated:

- - Can HMW turn the library seating chart into a live digital map?
- - HMW allowed students to "hold" a seat while walking from the bus stop?
- - How can HMW gamify the search for underused study areas?

5.3 Ideate Evidence (Youssef)

- Brainstorm ideas

List of Brainstormed Ideas (Divergent Phase):

- - The "SpaceSync" App (Chosen): Pressure sensors or cameras must be installed that display green/red dots for every table position on a map.
- - "Human Waze": Students estimate the number of people in the room in exchange for reward points in the cafeteria.
- - Rental Pods: Install soundproof rentable pods in the hallways (Rejected: Too Expensive for Students).
- - Smart tables. These are tables that light up physically when available. Rejected: "Doesn't solve the problem of walking there to check." - Drone Scout: A camera drone that flies around the library to count heads. (Rejected: Noise and privacy problems.)

5.4 Prototype Evidence (*Fares*)

Figure 1: Secure UTMID and Login Screen



Figure 2: Dashboard/Home screen where users can check general sensor analysis

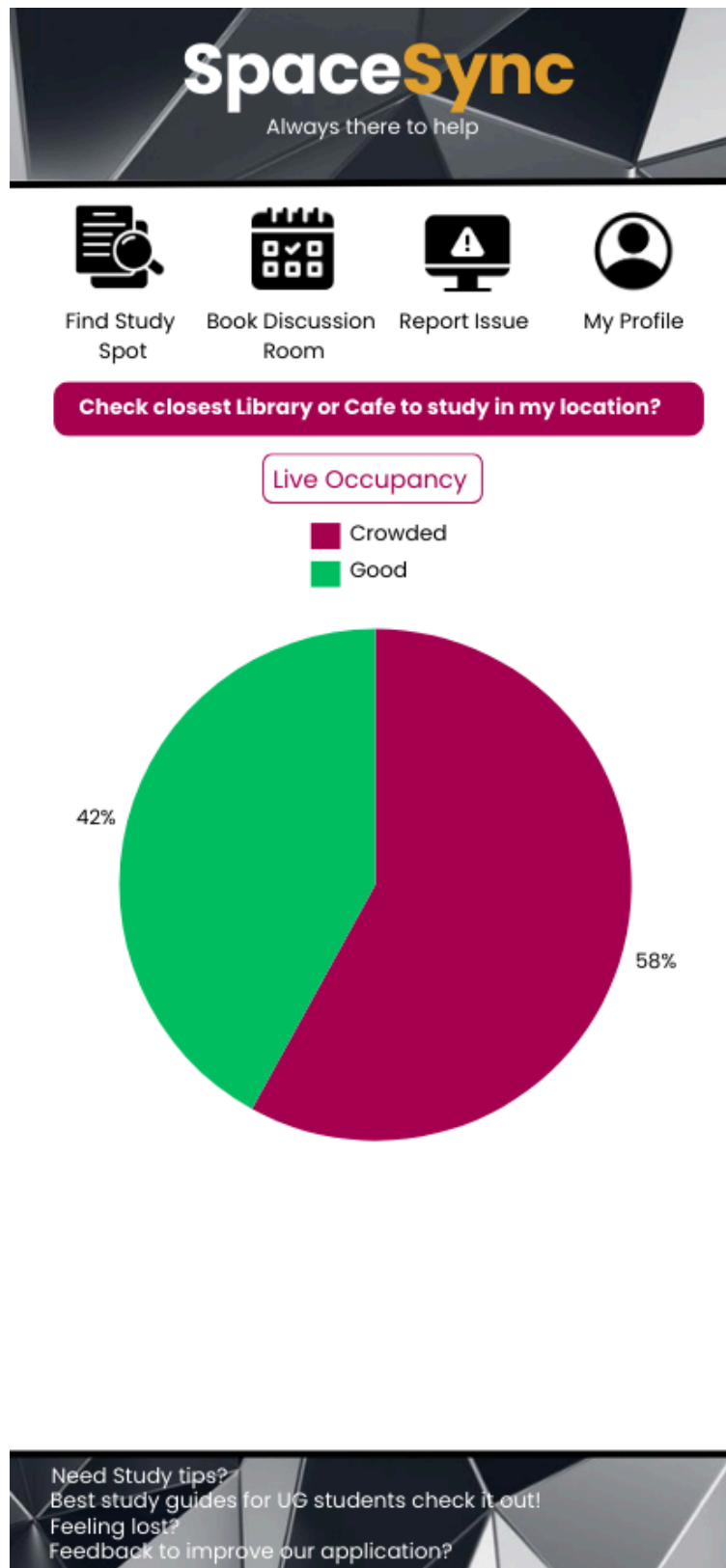
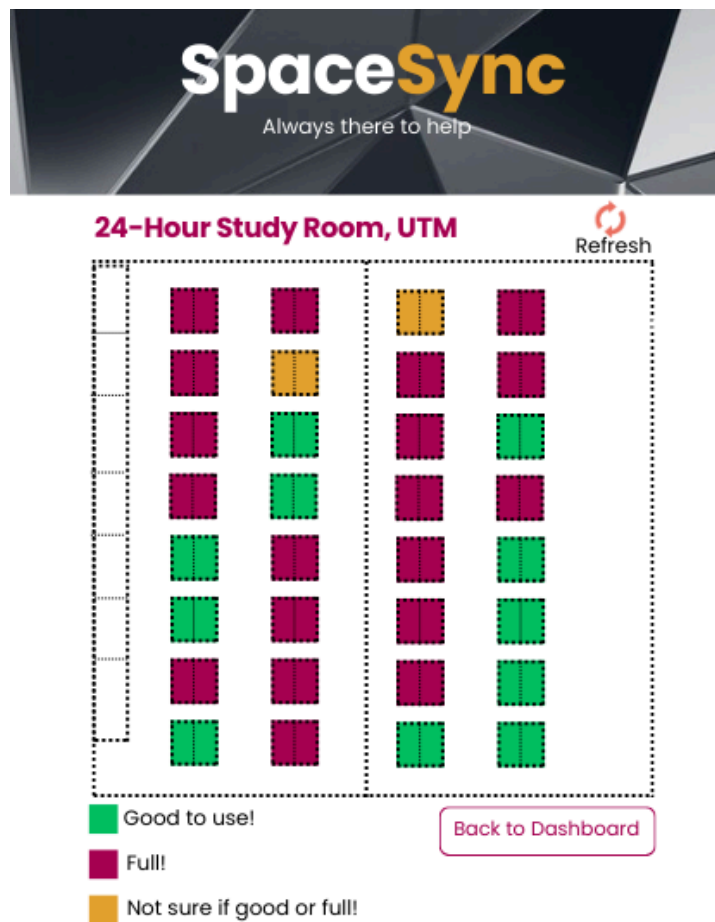


Figure 3: Screen showing actual real-time available and occupied tables



Need Study tips?
Best study guides for UG students check it out!
Feeling lost?
Feedback to improve our application?

Figure 4: Confirmation screen for booking verification

Booking Table was successful!

Please be patient if table still occupied
and choose another table for booking,
thanks you.



Back

5.5 Test Evidence (*Fares*)

Tester	Task Assigned	Feedback/Comment	Action Taken
Student	Checking design and overall features	The user noted the interface appeared minimalistic. They stated they would use the app upon release but might revert to physical searching if the app showed no availability.	Enhanced design elements by adding more features and focusing on specific details raised in the feedback.
Staff	Check the logic of the application	The logic was sound, but the user raised concerns about the algorithm's recommendation accuracy. They suggested the system should account for specific needs (e.g., group study vs. quiet study).	Proposed an AI recommendation algorithm where students can input their specific needs (e.g., subject, group size), allowing the system to suggest the most suitable table or room.
Staff	Checking related features with real scenario in library	The user validated the concept ("Not bad") but expressed concerns regarding the hardware feasibility of installing sensors in a real-world scenario. They also questioned how analysis data is managed.	Proposed complex algorithms and robust security protocols to ensure system scalability. We focused on a modular design concept to ensure updates can be deployed without disrupting the live system.

Tester	Task Assigned	Feedback/Comment	Action Taken
Student	Checking design and overall features	The user noted the interface appeared minimalistic. They stated they would use the app upon release but might revert to physical searching if the app showed no availability.	Add more design features and improve features and focus on details of the feedback
Staff	Check the logic of the application	She found it logical but bothered when knowing the system is actually not very logical when comes to solve real issues with table like she said “student always can find place even in hard times but issues is where the best recommended for each one”	Add recommendation algorithms in the system using AI tools where student can ask where to go and what his subject is so system will find quite most suited place for them to study maybe for group so find good table or condition
Staff	Checking related features with real scenario in library	“Not bad”, he says but in the real scenario it is hard to install the system since the technology is still starting and it is not available. how to manage the analysis page he said?	So we add more complex algorithms and uses better code and security features to protect the knowledge for UTM and keep it safe and use more suitable design concepts so it won't affect when we update the system anytime or add more features.

6. Individual Reflections:

6.1 Ali Isameldin Ali

The objective that I have in mind for pursuing this course and program is to equip myself with the skills and knowledge necessary to become a competent and creative technology professional who can design digital systems that address real-world problems. Through the SpaceSync project, I learned how Design Thinking transforms abstract problems into user-centered solutions by focusing on empathy, clarity, and practicality.

As the team leader, I was responsible for coordinating tasks, ensuring alignment between sections, and compiling the final report into a coherent structure. This role strengthened my leadership, communication, and decision-making skills. Overall, this project reinforced my interest in smart system design within institutional environments and prepared me to contribute meaningfully to future digital transformation initiatives.

6.2 Mashrur Rahman Arnob

In my role as a Software Engineer, I would like to write people-centered and more intuitive software that bridges the gap between technology and real life. The design thinking methodology and, more specifically, the Empathy Phase of the SpaceSync project have opened my eyes to the need to consider the feelings of end users, such as frustrated or sleep-deprived students, rather than simply fulfilling the technical requirements. The greatest algorithm will be worthless if it does not target a problem and a need.

6.3 Youssef Mostafa Badr

As a future IT specialist, I am interested in specializing in Systems Analysis. I have a personal goal of designing efficient systems that are not only optimized in code but also take a human-friendly perspective. I would like to design a system that makes life easier for people by introducing a technology that serves people's emotional needs. Impact of Design Thinking: The "Define and Ideate" stages of this project forced me to change how I go about problem-solving. Before this, I might have immediately begun coding or developing a database. The way in which I thought about Sarah's problem, from finding a chair to reducing anxiety, has led me to understand how having an excellent technical solution (sensors) makes no sense if one does not also have an effective user interface (live map), one that provides psychological comfort.

Improvement Plan:

To increase my possible skill level in the area, I hope to integrate these concepts directly in the

work I do with the technical skills I have now. I would like to learn about the types of tools available to integrate the knowledge of data with the art of the user experience—think data visualization libraries and handling data in real time (think Firebase), for example. “Reframing” issues in everyday activities is also about being sure to address the underlying issue rather than just treating the symptom.

6.4 Fares Hamed

As a software engineer, I am trying to achieve a personal goal of becoming competent in all skill sets that are applicable to the marketplace. I want to be at least an elite-level professional in this field, with the ability to compete at that level and bring pride to my team. I believe I can use this knowledge to solve real-world problems with improved software.

The Design Thinking project offered me crucial knowledge and so much more. The real-world interviews that we had for the project increased my confidence level concerning my transition to becoming a software engineer. I also have plans to learn design principles in order to perhaps create apps myself in the future. Knowing how to design and get feedback before coding is imperative. I will also continue learning more about Canva to create a strong connection between design applications and implementation.

7. Task Distribution:

Ali:	Introduction, problem, solution, compilation
Mashrur:	Empathy (interviews, persona, insights)
Youssef:	Define & Ideate (problem statements, brainstorming)
Fares:	Prototype & Testing (screens, flow, feedback)